



FHV



DMT Laboratory Exercise 1: Monostables, Astables

Report

Fachhochschule Vorarlberg

Mechatronik MEB 4

Laboratory Exercise 1

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2 Abstract

The topic of the first laboratory exercise was monostable and astable multivibrators. The goal was to analyse the behaviour of multivibrators and compare the results with the theoretical knowledge as well as with the calculated results. Due to the influence of the impedance of the used measurement unit pulse times can vary from the calculated ideal times. Additionally, the influence of fast switching operations of multivibrators on the voltage supply were analysed. It should also be examined how a astable multivibrator could be built by using two CMOS inverters. As well as what is the influence of the voltage supply level on quartz oscillator circuits.

3 Monostable Multivibrators

3.1 Introduction

In the first task of the Laboratory Exercise it was asked to build a Monostable Multivibrator and measure the pulse duration of the output signal as well as the voltage across the capacitor. This measurement should be carried out by using test probes as well as using standard BNC cables.

To build the Monostable Multivibrator a NE555 Timer IC (integrated circuit) has been used, in the Figure 2 the simplified structure this IC is shown. It contains of three resistors with the same value, two comparators, a Flip-Flop, a Bipolar Transistor and an Inverter at the output.

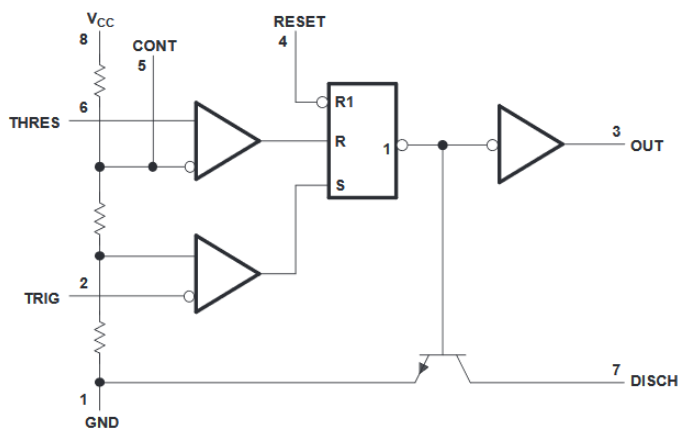


Figure 1: simplified structure of the NE555

The wiring of the NE555 was carried out like shown in Figure 2.

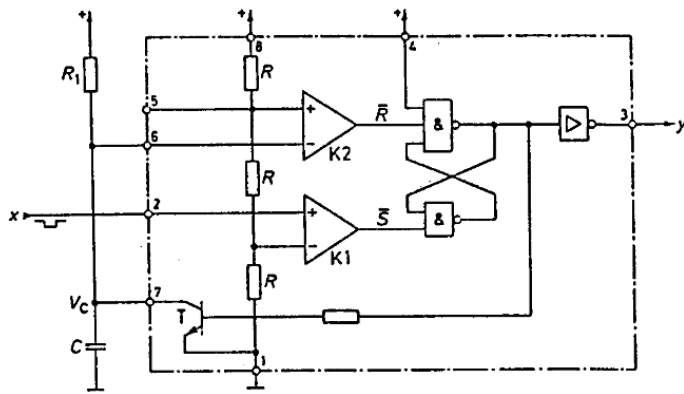


Figure 2: Wiring of the NE555 to get a Monostable Multivibrator

The expected result for the experiment was that after the Trigger Input gets a falling edge the output of the Multivibrator is high until the charging voltage of the capacitor reaches two-thirds of the operating voltage. (Figure 3)

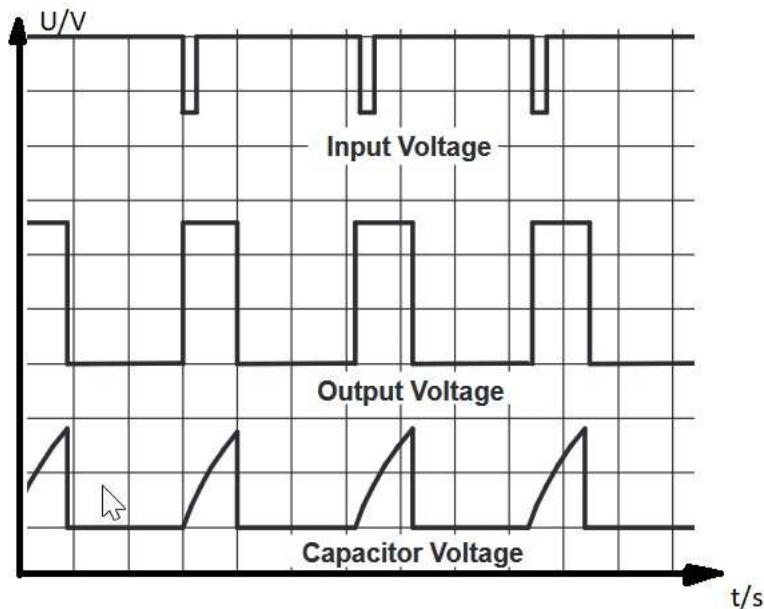


Figure 3: Schematic Signal behaviour of the Monostable Multivibrator

The voltage level of two-thirds of the operating voltage is a result of the two to one voltage divider at the Input of the upper comparator. As soon as the capacitor reached this voltage level the Transistor gets a high and shortens the capacitor. Which means that the Output of the Monostable Multivibrator is low because of the inverter.

3.2 Procedure

According to the instruction of the Laboratory Exercise the values of the used components are; $C=1\mu F$, $R=1M\Omega$, $V_{cc}=10V$. In order to get the falling edge for the Trigger the circuit shown in Figure 4 has been used.

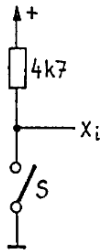


Figure 4: Trigger Button

For the measurement of the output and the capacitor voltage a four-channel Oscilloscope (TELEDYNE LECROY HDO6104-MS) with an input impedance of $1M\Omega$ has been used. The used test probes had a 10:1 ratio.

3.3 Results and Discussion

In the preparation for the laboratory exercise the pulse duration of the output signal should be calculated. For the derivation of the charging time of the capacitor the collector-emitter saturation voltage of the transistor has also been taken into account. This calculation is shown in Figure 5 below.

$$U_C(t) = V_{CC} * \left(1 - e^{-\frac{t}{RC}}\right)$$

$$t = \ln\left(\frac{V_{CC}}{V_{CC} - U_C(t)}\right) * RC$$

$$t_{0-0.3V} = \ln\left(\frac{10}{10 - 0.3}\right) * 1\mu F * 1M\Omega = 0.03s$$

$$t_{0-6.667V} = \ln\left(\frac{10}{10 - 6.667}\right) * 1\mu F * 1M\Omega = 1.1s$$

$$t_{0.3-6.667V} = t_{0-6.667V} - t_{0-0.3V} = \mathbf{1.07s}$$

In the first measurement both the output voltage as well as the capacitor voltage should be measured by using the test probes. The measurement setup is shown in the Figure 5.

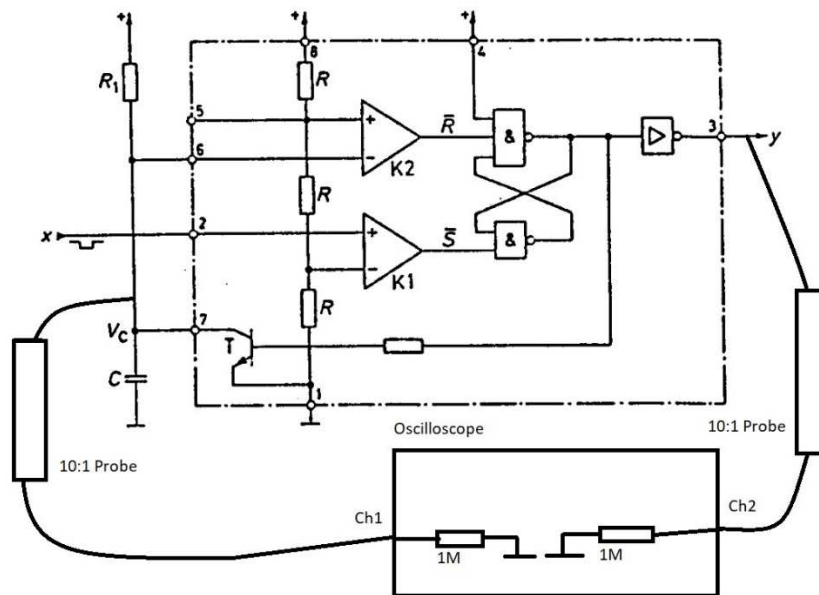


Figure 5: measurement setup with test probes

The measured result for the pulse duration was **1.17s**. This is 9.3% longer than the calculated result. This can be explained by the $10\text{M}\Omega$ Resistor of the measurement setup. Because of the fact that the R_1 resistor is $1\text{M}\Omega$ the input impedance of the Oscilloscope has a big effect on the charging time. The voltage which the capacitor gets to charge is in this case calculated like shown below:

$$U_c = V_{cc} * \frac{10\text{M}\Omega}{1\text{M}\Omega + 10\text{M}\Omega} = 9.09\text{V}$$

This lower charging voltage leads to the result that the charging time until the capacitor reaches the voltage of two-thirds of the operating voltage rises. In the Figure 6, the measured graph is shown.

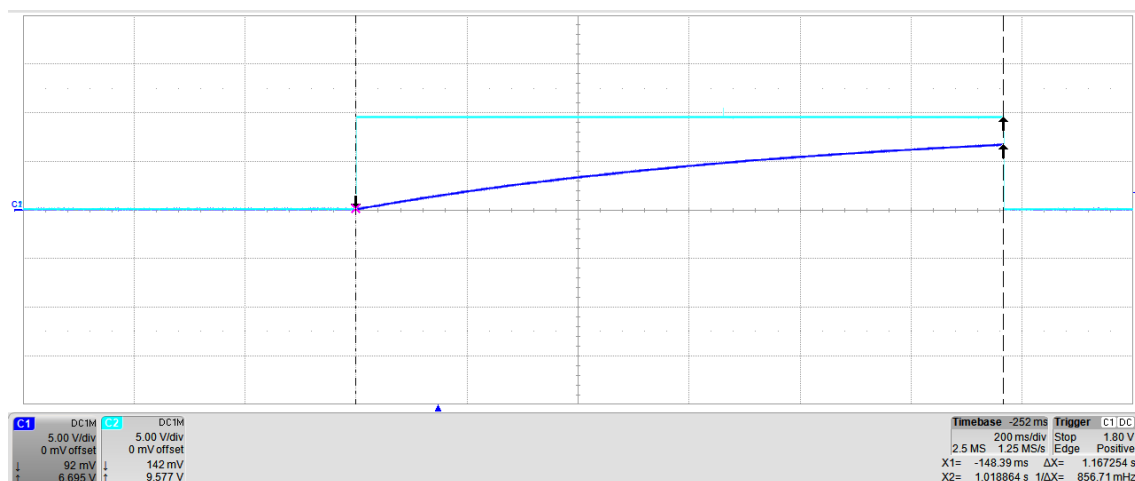


Figure 6: Output and charging voltage measured with test probes

The next measurement was performed by using the standard BNC cables. Therefore, the measurement setup was like shown in the Figure 7 below.

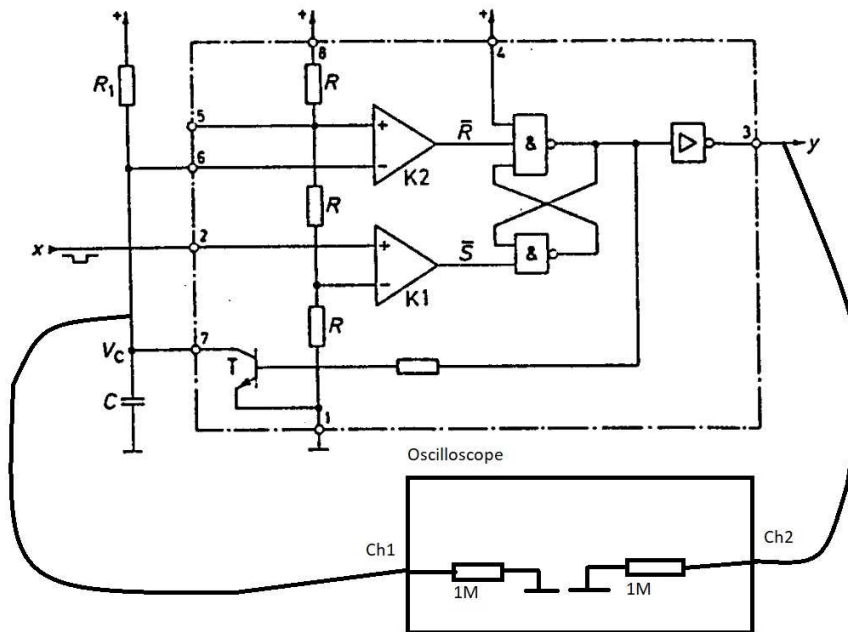


Figure 7: measurement setup with BNC cables

The measured result in this case was that the output was always high because the capacitor never reached the two-thirds of the operating voltage, this can be seen in Figure 8.

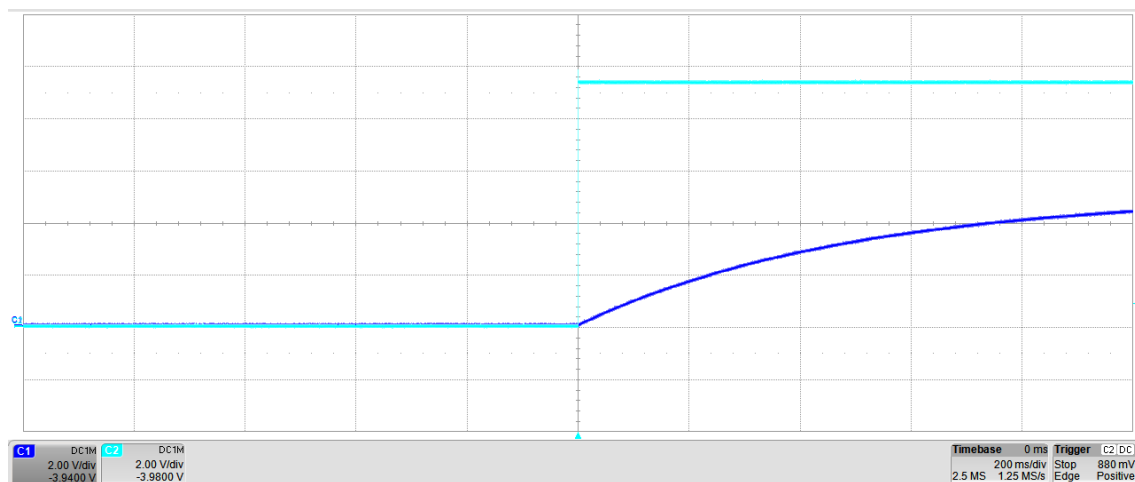


Figure 8: Output and charging voltage measured with BNC cables

The reason for this behaviour was that due to the fact that no test probes were used, the input impedance of the measurement circuit was $1\text{M}\Omega$, which is the same value like the R_1 resistor. This caused the issue, that the charging voltage for the capacitor dropped to 50% of V_{cc} . This means the capacitor was never charged to two-thirds of V_{cc} . Therefore the output was always high.

In the next step there was only the output voltage measured without measuring the capacitor. In this case the pulse time of the output signal was at **1.07s**. This is exactly the calculated value. In Figure 9 this behaviour is visualized.

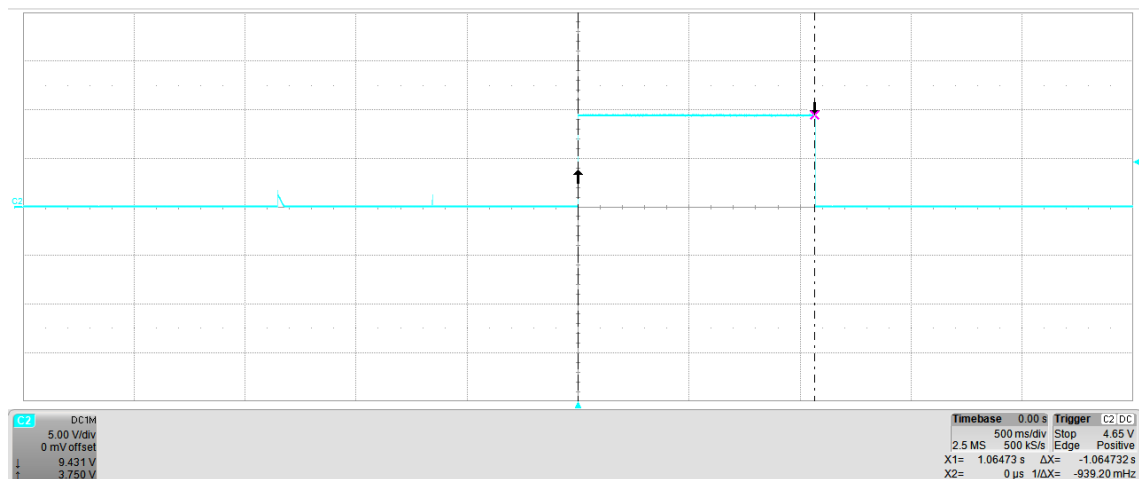


Figure 9: Output measured with BNC cable

The reason for this exact pulse duration is that due to the fact that the capacitor had no resistor parallel, the charging voltage was V_{cc} .

3.4 Conclusion

If the pulse duration of the output signal of a Monostable Multivibrator should be measured it's best to measure only at the output because otherwise the impedance of the measurement instrument can affect the pulse duration.

4 Astable Multivibrators

4.1 Introduction

The second task for the laboratory exercise was to build a astable multivibrator and also measure the output pulse duration and the voltage across the capacitor. The astable multivibrator should also be built by using the Timer IC NE555, the wiring of the astable multivibrator was carried out like shown in the Figure 10.

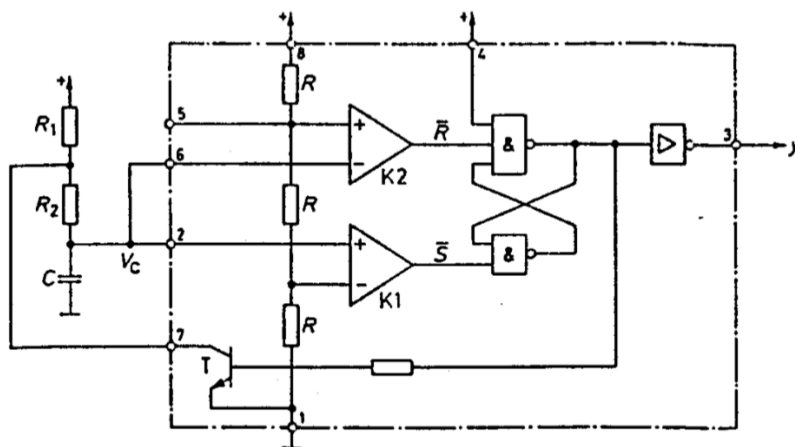


Figure 10: Wiring of the astable multivibrator

The expected result of the measurement is that as soon as the operation voltage is applied to the circuit the astable multivibrator generates a rectangular voltage at the output. The high

time at the output is determined by the time which the capacitor needs until it's charged from one-third of V_{CC} to two-thirds of V_{CC} and the low time is the time which the capacitor needs to get discharged from two-thirds V_{CC} to one-third V_{CC} . In Figure 11 this behaviour is shown.

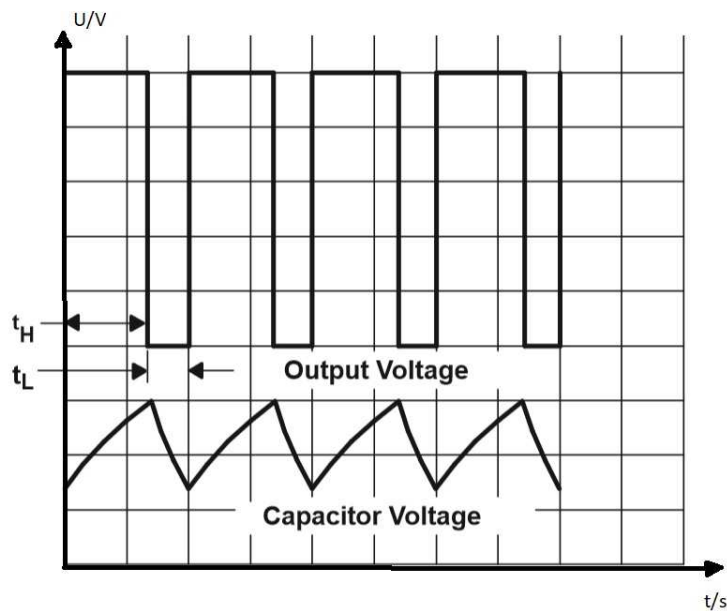


Figure 11: schematic behaviour of the astable multivibrator

4.2 Procedure

According to the instructions of the laboratory exercise the following components had to be used; $R_1=R_2=10k\Omega$, $C=100nF$, $V_{CC}=10V$. For the measurement of the output and the capacitor voltage a four-channel Oscilloscope (TELEDYNE LECROY HDO6104-MS) with an input impedance of $1M\Omega$ has been used. The used test probes had a 10:1 ratio

4.3 Results and Discussion

As preparation for the laboratory exercise the vibration period should be calculated. This calculation is visible below.

High Time:

$$t = \ln\left(\frac{V_{CC}}{V_{CC} - U_C(t)}\right) * RC$$

$$t_{0-3.333V} = \ln\left(\frac{10}{10 - 3.333}\right) * 100nF * 20k\Omega = 0.81ms$$

$$t_{0-6.667V} = \ln\left(\frac{10}{10 - 6.667}\right) * 1nF * 20k\Omega = 2.2ms$$

$$t_{3.333-6.667V} = t_{0-6.667V} - t_{0-3.333V} = \mathbf{1.39ms}$$

Low Time:

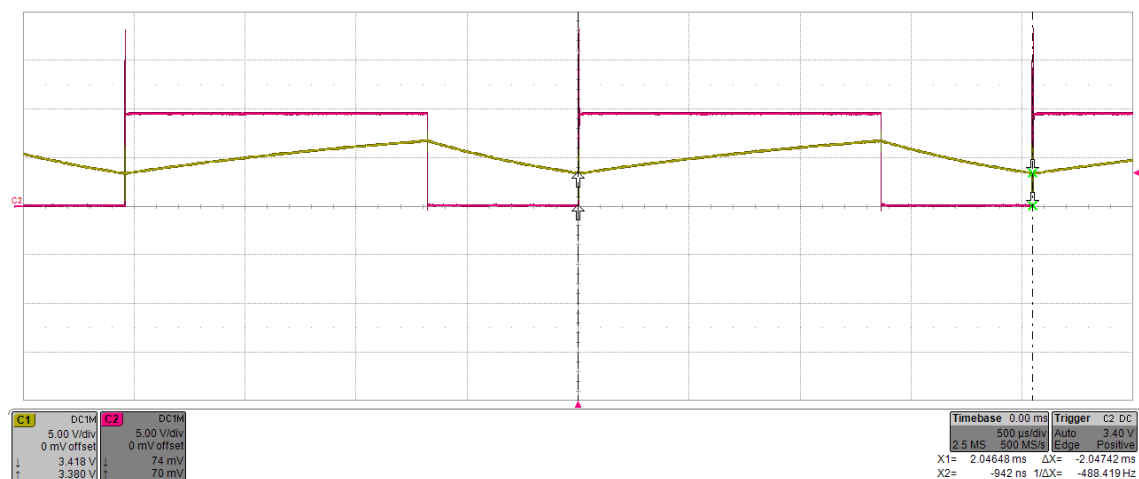
$$t = \ln\left(\frac{V_{CC}}{U_C(t)}\right) * RC$$

$$t_{6.667-3.333V} = \ln\left(\frac{10}{3.333}\right) * 100nF * 10k\Omega = \mathbf{0.69ms}$$

$$T = t_{3.333-6.667V} + t_{6.667-3.333V} = 2.08ms$$

The circuit diagram shows a monostable multivibrator using two 555 timers, K1 and K2. The power supply is V_C . The timing network consists of a capacitor C and resistors R_1 , R_2 , and three resistors R . The output y is connected to an oscilloscope via a 10:1 probe. The oscilloscope shows a single rectangular pulse on Ch1 and a corresponding pulse on Ch2.

The measured vibration time was **2.08ms** which equals a frequency of about 400Hz. This measurement graph is shown in the Figure 13 below. Due to the fact that the input impedance of the oscilloscope is much higher than the resistors used in the circuit the deviation from the measured vibration time to the calculated one is negligibly small.



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For the next step it has been asked to analyse the rising edge of the output signal (Figure 14).

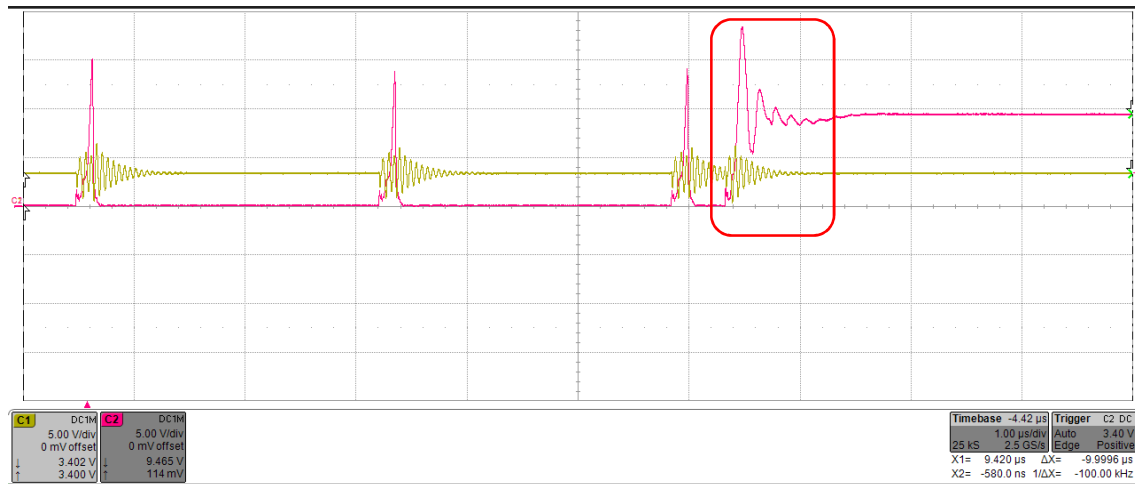


Figure 14: Rising edge of the output signal

The reason for the oscillation at the rising edge of the output signal is due to the fact, that there was no decoupling capacitor built into the circuit at first. This one is needed because if there is a fast switching happening on the output the capacitor provides the current in this small time gap until the voltage supply catches up. This behaviour is shown in the graph "Figure 15" below. During this measurement a 10nF decoupling capacitor was used.

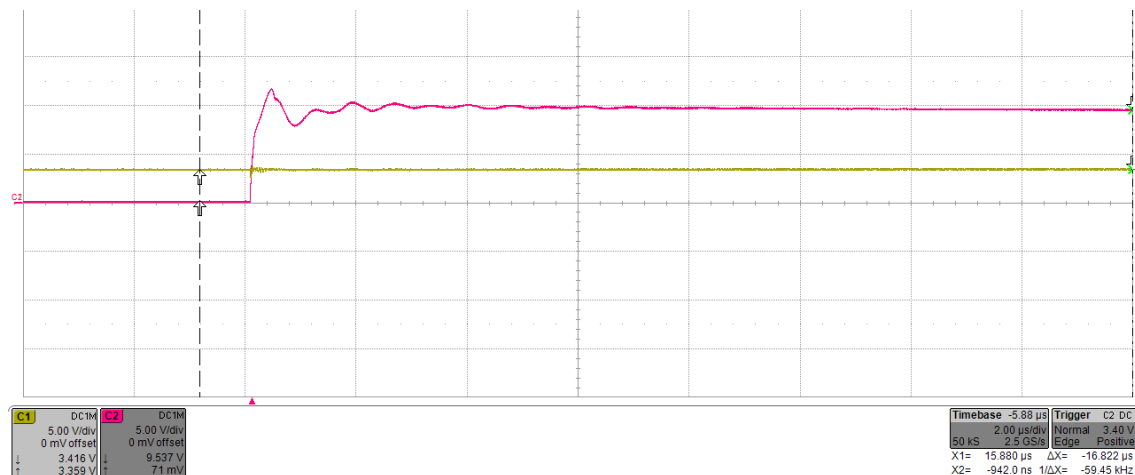


Figure 15: rising edge with 10nF capacitor

4.4 Conclusion

In order to ensure a stable voltage supply for ICs during fast switching operations a capacitor has to be connected parallel to the supply voltage. This capacitor functions as energy storage and supplies the IC for a short period of time. It has to be connected as near as possible to the IC to keep the parasitic inductivity as low as possible.

5 Astable Multivibrator Using Two Inverter

5.1 Introduction

The task of this experiment is to understand the function of the circuit of the figure Astable Multivibrator. To get a clue what the circuit does, the output voltage and the voltage at point P should be measured with the oscilloscope. Over there the voltage curves on the oscilloscope at the points P and S should be explained and what the risen for the resistor R_S is.

5.2 Procedures

The first step is to assemble the circuit on pinboard. For the two Inverter we use the integrated CMOS NAND-circuit 74C00. Therefore, the inputs of one NAND should be connected. We use 10V as supply voltage for it. The other elements we use are the two resistors $R = 10\text{k}\Omega$ and $R_S = 1\text{k}\Omega$ and the capacitor $C = 1\text{nF}$. For measurement with the oscilloscope we use probes. There it is important that the ground at each measurement is at the same potential.

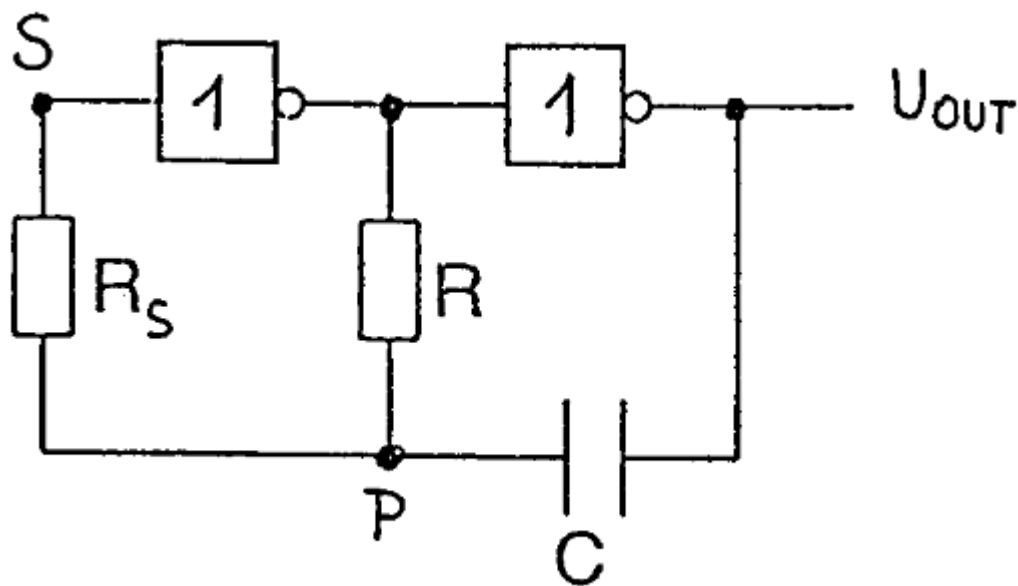


Figure 16: Astable Multivibrator

5.3 Results and Discussion

To understand the function of the circuit we need the graph from the oscilloscope that is imaged below.



Figure 17: Astable Multivibrator voltage on Out and P

On channel 1 you see the voltage from the capacitor and on channel 2 you see the output voltage form astable multivibrator. Like expected from an astable multivibrator the output is oscillating between 0V and 10V. To explain the function, you start at the state when the voltage from out at P is equal 0V. The Input of 74C00 has a high protection resistor in relation to the Rs. Therefore, the current through Rs is very low and the Point P and S has nearly the same voltage. The left inverter interprets this as a low and produce between the two inverter (point Q) a logic high and this is with our supply voltage equal to 10V. With this input the right inverter produce at the output 0V. The potential between Q and Uout generate a current through the capacitor is getting charged. The voltage from P and S is rising until the Inverter interpret the input as a logic high. From the oscilloscope you see that this point must be around $0.6 \cdot V_{cc}$. Q will go to 0V and Uout to 10V. Because the capacitor determines a potential difference the voltage P is going to $10V + 0.6 \cdot V_{cc} = 16V$. At the figure you see only 14,6V. Probably two protection diodes from the inverter make the different between the voltages. In this state the current flow in the other direction and the capacitor is discharging until the voltage S is under 6V. The inverters change the state and the voltage in point P should be 10V lower. Here you could see also the 1,4V different for the diodes.

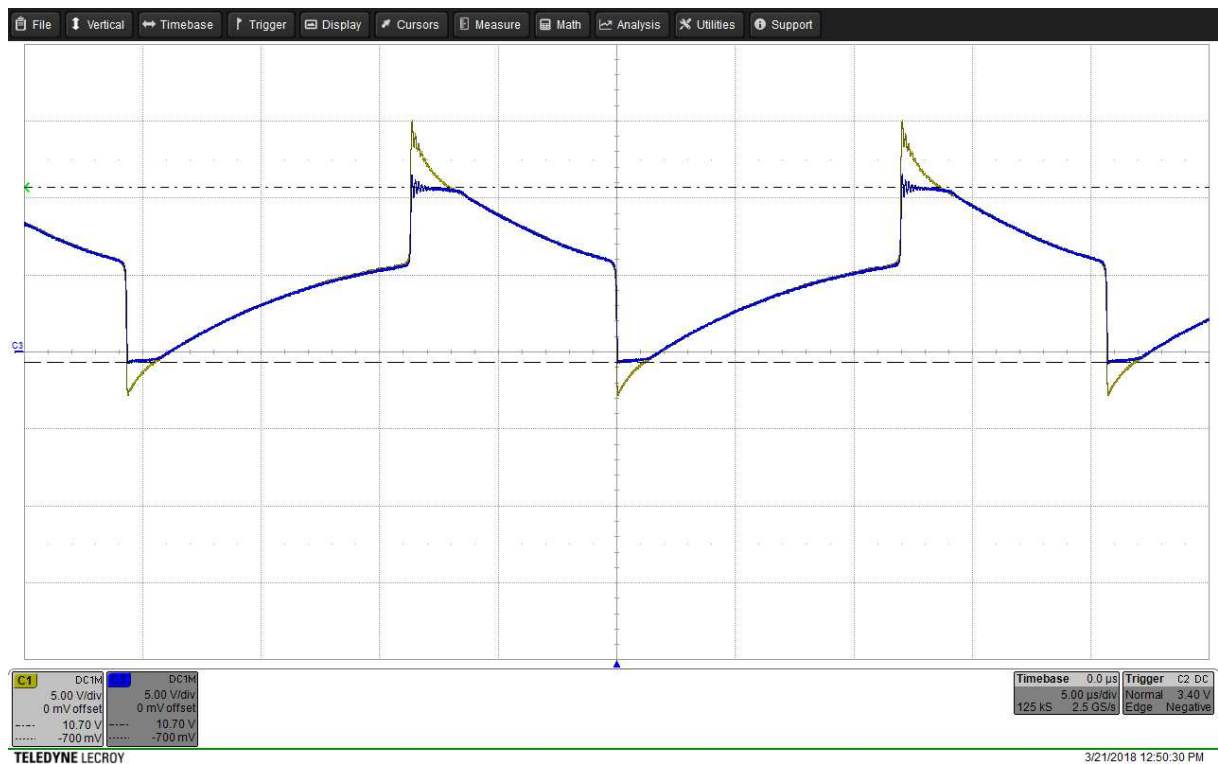


Figure 18: Inverter Input Voltage

On channel 1 you see the voltage from the capacitor and on channel 2 you see the input voltage for the left inverter. The two curves are nearly the same. Only the high and low peak is cut off. That comes from the inverter protection diodes. The voltage is cut off the diodes very fast, but because of the high inverter protection resistor the current is very low. The discharge curve for the capacitor is depending on the current through it. Thus, the diodes nearly don't change the oscillation time. The resistor R_s isn't need for function because the inverter has anyway a resistor. R_s is only used the see better how the circuit works.

5.4 Conclusions

With two inverters and one resistor and capacitor you could built up a very easy astable multivibrator. Until the Inverters change the state the capacitor saves the charge and the charge curves determine the oscillation time.

6 Quartz Oscillator

6.1 Introduction

The task of this experiment is measure and explain the curves from the quartz oscillator with the oscilloscope.

6.2 Procedures

The first step is to assemble the circuit on pinboard. For the two Inverter we use the integrated CMOS NAND-circuit 74C00 and the faster 74HC00. Therefor the inputs of one NAND should be connected. We use 5V as supply voltage for it. The other elements we use are the two resistors $R1 = 1\text{M}\Omega$ and $R2 = 220\ \Omega$, the capacitors $C1 = C2 = 68\text{nF}$ and for oscillation a 12MHz quartz crystal. For measurement with the oscilloscope we use probes.

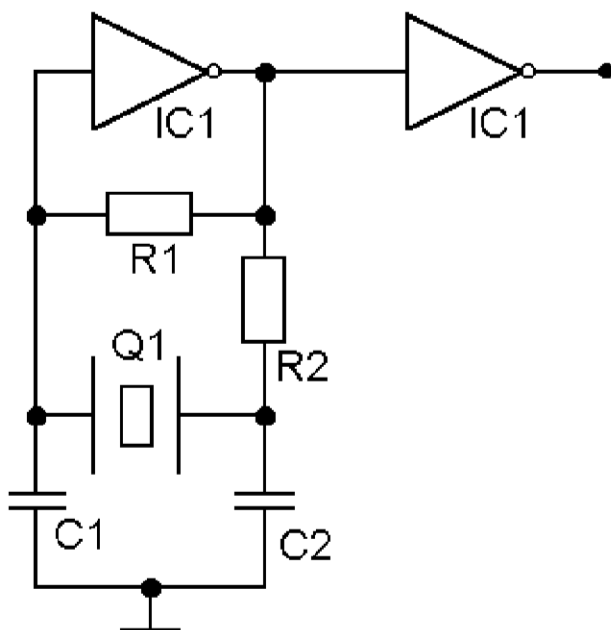


Figure 19 : Quartz Oscillator

6.3 Results and Discussion

With the 5V Vcc you get on the oscilloscope a constant 4,28V value. That's not what we expect from an oscillator. The reason is the propagation delay time for logic 0 or 1, who is dependent on the supply voltage. This value at 5V Vcc is typical 50ns and maximum 90ns. That mean the 74C00 circuit isn't typical faster than 100ns for one oscillation. This 12Mhz quartz need a period time form 83ns.

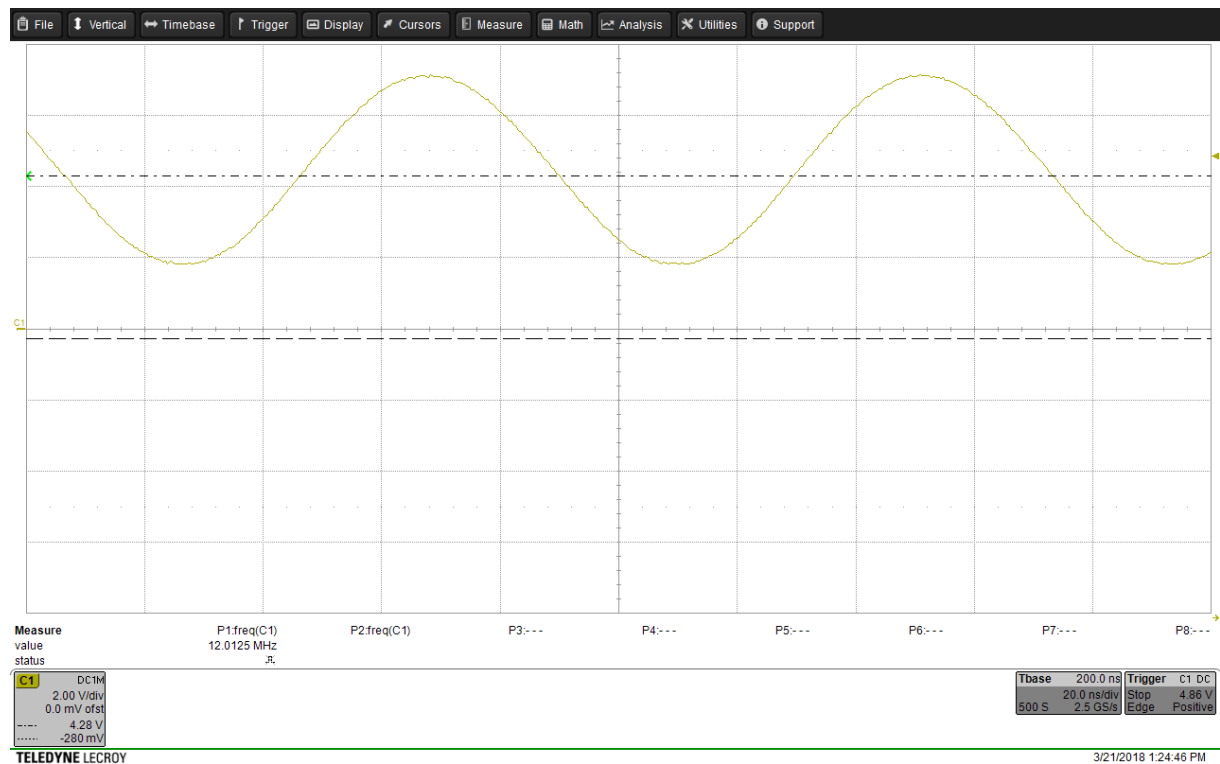


Figure 20: Quartz Oscillator with 10V Vcc

With the 10V power supply you see on the oscilloscope a proper sinus curve. The typical preparation delay time from the datasheet with 10V Vcc is 30ns. That would be fast enough for this quartz.

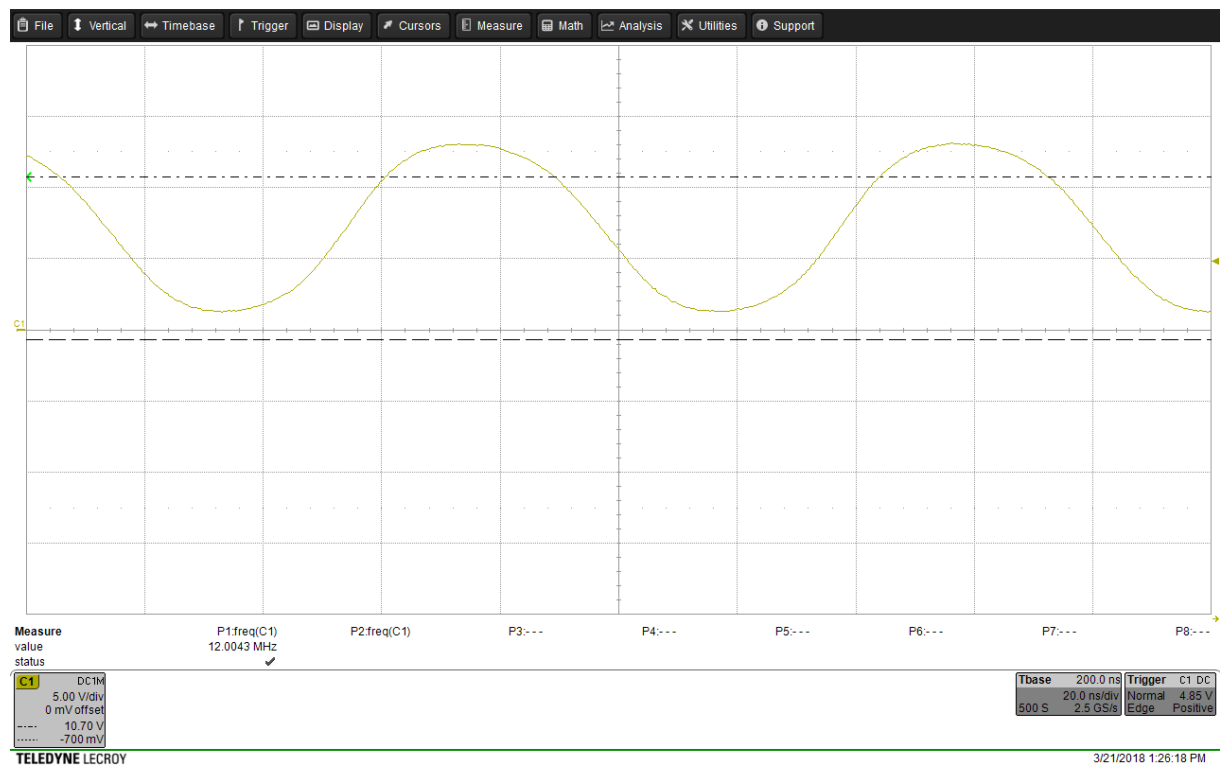


Figure 21: Quartz Oscillator with 15V Vcc

With the 15V Vcc you see a distortion because the propagation delay time is faster than the quartz.

6.4 Conclusions

If you implement a quartz oscillator you have to check the propagation delay time and the power supply from the datasheet. The values from the period times should fit together. Otherwise the oscillator couldn't work properly.

7 References

- Data Sheet Texas Instruments xx555 Precision Timers
- DMT Laboratory Exercise 1: Monostables, Astables
- Data Sheet Texas Instruments 74C00
- Data Sheet Texas Instruments 74HC00
- Lab Template Lab Report Guide